

Electroencephalographic Assessment of Comfort in Urban Environments Using Virtual Reality

Yasufumi Nakata^{1*}Mayuka Otsuki^{1,2†}Taisuke Goto^{1§}Souta Hiratsuka^{1¶}Miki Nagai^{1||}Kaito Sado^{1**}Atsushi Aoyama^{1††}¹ Keio University² Oberlin College

ABSTRACT

Rapid urbanization affects well-being and stress, motivating research in “neurourbanism”. However, many studies rely on 2D representations of urban environments, which lack realism. We combined VR with 64-channel electroencephalography to assess comfort in urban environments. Green visibility induced reduced heart rate, suggesting relaxation. Building height induced increased beta-band activity, suggesting cognitive engagement. Building coverage most strongly affected the perceived restorativeness scale, suggesting subjective restoration. This multi-indicator approach enabled us to pinpoint specific urban elements linked to different aspects of comfort, demonstrating that VR combined with diverse physiological and subjective measures can effectively identify comfort-related factors.

Index Terms: Neurourbanism, Urban environment, Comfort, Electroencephalography (EEG), Electrocardiography (ECG).

1 INTRODUCTION

Urban environments influence people’s well-being and stress. The emerging field of “neurourbanism” addresses this using physiological measures such as electroencephalography (EEG) and autonomic metrics to guide evidence-based urban planning [1, 2, 3]. To present urban environments, previous studies have typically used static 2D images and videos [4, 5]. However, they fail to fully explain urban comfort mechanisms due to the lack of realism in 2D representation. Additionally, existing research lacks systematic control over multiple environmental factors and shows insufficient integration of subjective and physiological evaluations.

To address these limitations, this study utilizes VR to systematically manipulate four environmental factors—building coverage, overall building height, green visibility ratio, and noise—while evaluating comfort via EEG, electrocardiography (ECG) [1, 2, 3], and perceived restorativeness scale (PRS) [6]. We hypothesize that immersive VR allows realistically controlled urban simulations. Furthermore, we propose that combining physiological markers (EEG and ECG) with PRS will enable a quantitative assessment of comfort and allow us to identify which specific urban elements contribute to different aspects of comfort. This multi-indicator approach aims to resolve prior methodological limitations by enabling

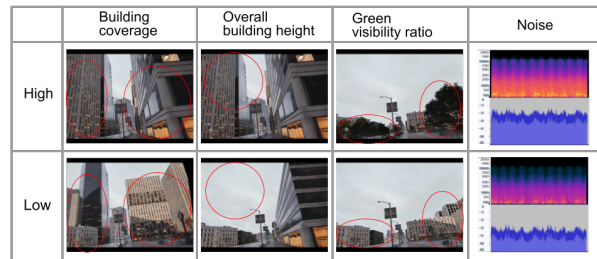


Figure 1: VR stimuli for four urban parameters.

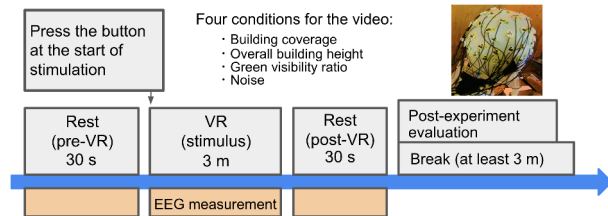


Figure 2: Overview of experiment.

us to pinpoint which urban elements are linked to comfort-related responses.

2 MATERIALS AND METHODS

2.1 Experimental Procedure

Sixteen participants (nine females, mean age 20.6 ± 1.4 years) participated in the experiment. The experimental protocol was approved by the Keio University SFC Research Ethics Committee.

Participants experienced VR content using Meta Quest 3 (Meta Inc.) while seated. The VR content was created using Unreal Engine 5 (Epic Games Inc.), reproducing urban environments through stereoscopic VR. Participants passively viewed VR content showing movement through urban spaces, with four factors varying between high and low levels: (1) building coverage (Coverage), (2) overall building height (Height), (3) green visibility ratio (Green Visibility), and (4) noise (Noise). Each participant performed 16 trials, viewing all combinations ($2 \times 2 \times 2 \times 2 = 16$ conditions) in a randomized order (Fig. 1). Each trial consisted of pre-VR (30 s), VR stimulus (3 m), and post-VR intervals (30 s) (Fig. 2).

2.2 Data Collection and Analysis

EEG and ECG data were recorded with a 64-channel actiChamp system (Brain Products GmbH), with ECG measured using wrist-ankle derivation. Subjective restoration was evaluated using PRS after each viewing.

EEG data were preprocessed using EEGLAB: resampling at 500 Hz, removing noise (50/60 Hz and harmonics), bandpass filtering (1–100 Hz), bad channel detection (kurtosis-based, ± 3.5 standard deviation threshold), spherical spline interpolation, and automatic

*e-mail: yasufumi@keio.jp

†e-mail: mayukaotsuki314@keio.jp

‡e-mail: motsuki@oberlin.edu

§e-mail: goto.taisuke@keio.jp

¶e-mail: souta.hiratsuka@keio.jp

||e-mail: mikinagai0920@keio.jp

**e-mail: ksza@keio.jp

††e-mail: aaoyama@sfc.keio.ac.jp

Table 1: Summary of findings (C: Coverage, H: Height, G: Green Visibility, and N: Noise).

Modality	Outcome	Factor(s)	Statistics
EEG	Beta-band PSD difference	H	$F = 3.67, p = 0.075$
EEG	Theta-band PSD difference	H×G	$F = 3.89, p = 0.067$
EEG	Beta-band PSD difference	C×H×G×N	$F = 3.53, p = 0.080$
EEG	Gamma-band PSD difference	H×G×N	$F = 3.85, p = 0.069$
ECG	HR in VR	G	$F = 5.77, p = 0.037$
ECG	HR difference	G	$F = 4.35, p = 0.019$
PRS	Total score	C	$F = 4.57, p = 0.051$

artifact removal. Power spectral density (PSD) in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz) frequency bands was calculated from posterior and frontal EEG electrodes (O1, O2, F3, and F4). These electrodes were selected based on established protocols: posterior electrodes (O1 and O2) for relaxation assessment [7], and frontal electrodes (F3 and F4) for attention and cognitive load evaluation [8]. Heart rate (HR) was computed based on R-peaks detected from ECG signals. For statistical analysis, repeated-measures analysis of variance was performed on each measure with the four factors.

3 RESULTS

The statistical results are summarized in Table 1. Statistical analysis of the difference of PSD between pre-VR and post-VR intervals revealed no statistically significant main effects or interactions in any band at the conventional alpha level. However, several marginally significant effects were observed: a main effect of Height in the beta band ($F = 3.67, p = 0.075$), with PSD values 0.53 dB higher under the high-Height condition than under the low-Height condition; a two-way interaction between Height and Green Visibility in the theta band ($F = 3.89, p = 0.067$); a three-way interaction among Skyline, Green Visibility, and Noise in the gamma band ($F = 3.85, p = 0.069$); and a four-way interaction among Coverage, Height, Green Visibility, and Noise in the beta band ($F = 3.53, p = 0.080$).

As for the HR during the VR viewing period, Green Visibility emerged as the only factor showing a significant main effect ($F = 5.77, p = 0.037$). Specifically, the high Green Visibility condition yielded a mean HR reduction of 1.94 bpm relative to the low Green Visibility condition. Moreover, the difference of HR between pre-VR and post-VR intervals revealed a significant main effect of Green Visibility ($F = 4.35, p = 0.019$), with a 2.83 bpm reduction under the high Green Visibility condition relative to the low Green Visibility condition.

Regarding PRS data from fifteen participants who completed all questionnaires, the main effect of Coverage was almost significant ($F = 4.57, p = 0.051$). Furthermore, separate analyses for each PRS subscale factor (Being Away, Fascination, Coherence, Scope, Compatibility, Preference, and Familiarity) revealed significant main effects for Fascination ($F = 5.98, p = 0.028$), Scope ($F = 5.52, p = 0.034$), Compatibility ($F = 5.67, p = 0.032$), and Preference ($F = 4.74, p = 0.047$).

4 DISCUSSION

The reduction of HR and the virtually constant beta- and gamma-band PSD under Green Visibility indicate that visual natural elements promote relaxation through the autonomic nervous system rather than the cognitive system. In contrast, an increase in beta-band PSD without concomitant changes in HR under Height indicates induced cognitive arousal, likely reflecting visual exploration and attentional mobilization prompted by vertical expansion and

complexity. These findings suggest moderate cognitive engagement rather than physiological stress and are consistent with the “fascination” process in the attention restoration theory [9], which characterizes built environments as cognitively demanding and natural settings as restorative. Moreover, the finding that Coverage and Noise were physiologically neutral, whereas Coverage was subjectively the most dominant, provides important insight for VR-based urban simulations. This component-specific differentiation suggests that urban design may benefit from optimizing distinct environmental features for rest versus vitality.

Limitations include the relatively small sample size ($N = 16$), which, given the high variability of physiological data, limits statistical power and places several key findings (e.g., beta power and total PRS) at the threshold of statistical significance. Some effects showed trend-level significance ($p \approx 0.05$) warranting verification in studies with larger samples. Additionally, the short duration of VR exposure (180 s) precludes assessment of long-term environmental effects.

5 CONCLUSION

By integrating VR with multiple indicators (EEG, ECG, and PRS), this study successfully identified which urban elements contribute to different aspects of comfort. We found that green visibility promotes physical relaxation (reduced HR), building height induces neural arousal (increased beta-band PSD), and building coverage most strongly affects subjective restoration (PRS). This multi-indicator approach demonstrates that VR combined with diverse physiological and subjective measures can effectively pinpoint specific urban elements linked to comfort-related responses, addressing the methodological limitations of previous research that relied on single indicators or 2D representations.

REFERENCES

- [1] M. Adli, M. Berger, E.-L. Brakemeier, L. Engel, J. Fingerhut, A. Gomez-Carrillo, R. Hehl, A. Heinz, J. H., N. Mehran, S. Tolaas, H. Walter, U. Weiland, and J. Stollmann. Neurourbanism: towards a new discipline. *The Lancet Psychiatry*, 4:183–185, 2017. 1
- [2] A. Olszewska-Guizzo, A. Sia, A. Fogel, and R. Ho. Can exposure to certain urban green spaces trigger frontal alpha asymmetry in the brain?—preliminary findings from a passive task EEG study. *Int J Environ Res Public Health*, 17(2):394, 2020. 1
- [3] A. Seiz, B.-S. Kweon, C. D. Ellis, H. Oh, and K. Pietro. Exploring the psychophysiological effects of viewing urban nature through virtual reality using electroencephalography and perceived restorativeness scale measures. *Sustainability*, 15(17):13090, 2023. 1
- [4] R. Reece, A. Bornioli, I. Bray, N. Newbutt, D. Satenstein, and C. Alford. Exposure to green, blue and historic environments and mental well-being: A comparison between virtual reality head-mounted display and flat screen exposure. *International Journal of Environmental Research and Public Health*, 19(15):9457, 2022. 1
- [5] J.-H. Chen, C.-H. Hsieh, S.-H. Hung, W. C. Sullivan, H.-F. Tang, and C.-Y. Chang. Using functional magnetic resonance imaging (fMRI) to analyze brain region activity when viewing landscapes. *Landscape and Urban Planning*, 162:137–144, 2017. 1
- [6] T. Hartig, K. Korpela, G. W. Evans, and T. Gärling. A measure of restorative quality in environments. *Scandinavian Housing and Planning Research*, 14(4):175–194, 1997. Taylor & Francis. 1
- [7] W. Klimesch. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. *Brain Research Reviews*, 29(2-3):169–195, 1999. 2
- [8] W. J. Ray and H. W. Cole. EEG alpha activity reflects attentional demands, and beta activity reflects emotional and cognitive processes. *Science*, 228(4700):750–752, 1985. 2
- [9] S. Kaplan. The restorative benefits of nature: Toward an integrative framework. *Journal of Environmental Psychology*, 15(3):169–182, 1995. 2